

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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D O P 3 1 1

Building Infrastructure as Code with AWS CloudFormation

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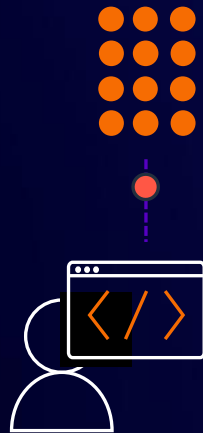
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Agenda

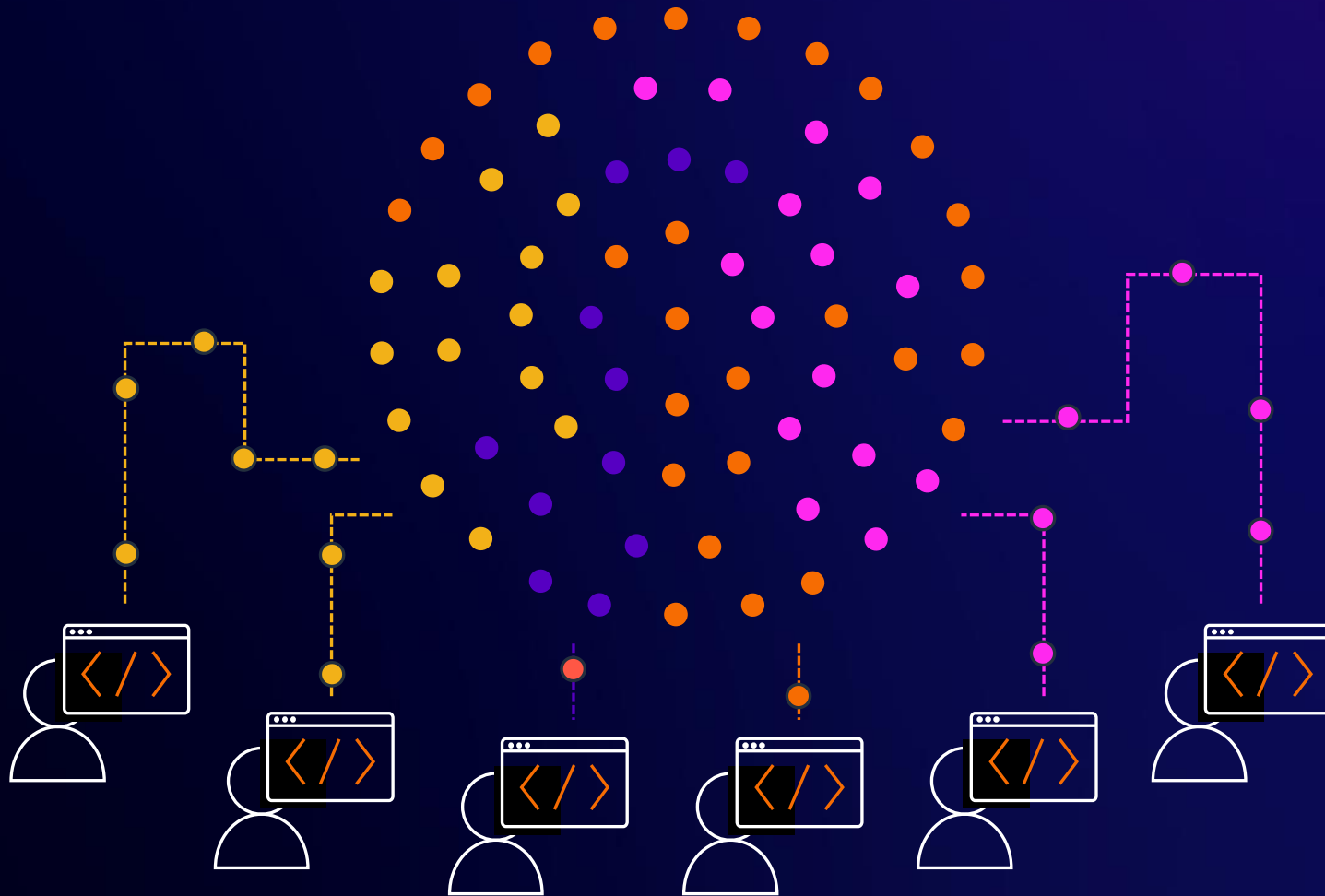
- What is Infrastructure as Code (IaC)?
- AWS CloudFormation essentials
- What is on your mind?

What is Infrastructure as Code?

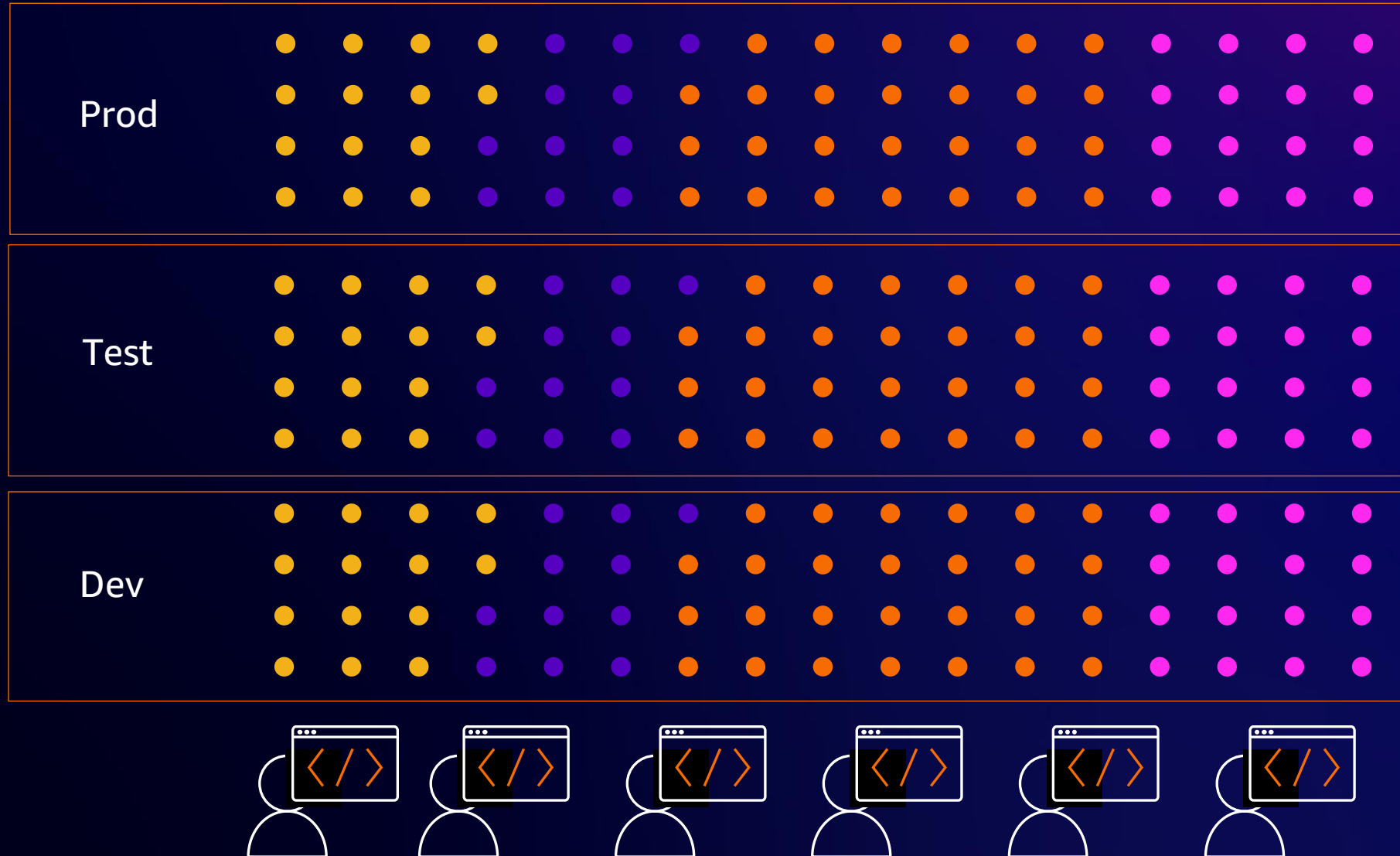
What is IaC?



What is IaC?



What is IaC?



How customers use AWS IaC

AWS IaC

Different authoring languages to meet customer choice to describe infrastructure

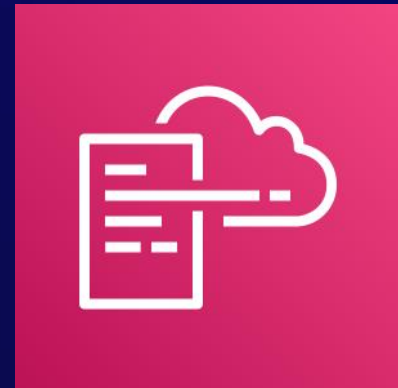
Automates provisions infrastructure as described in the template

Makes it easy to scale infrastructure

Tooling and integrations for enterprise requirements around compliance, governance, automation, and security

Manage AWS and third-party resources together

Over 1 million active customers manage application resources with AWS CloudFormation every week



Evolution of IaC

Abstractions

AWS CDK, Pulumi

Generators

Troposphere, GoFormation

Declarative

CloudFormation, Terraform

Scripted

`#!/bin/bash`

Manual

Wikis, playbooks, ask-Bob-he-knows

CloudFormation essentials

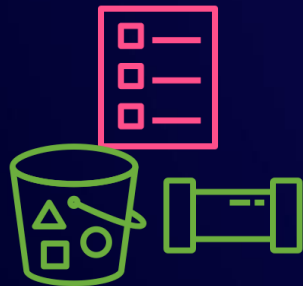


AWS CloudFormation at a glance



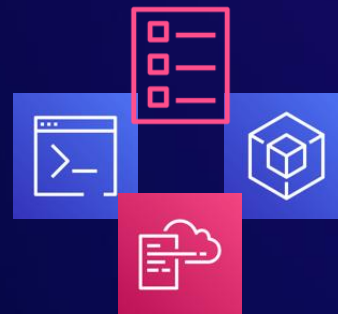
Code

Write templates in YAML or JSON, or generate code via macros (transforms) or higher-level languages (AWS CDK)



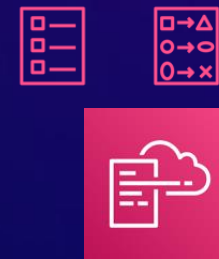
Commit

Upload templates directly via browser, from an Amazon S3 bucket, or ideally via CI/CD pipelines to leverage testing tools



Execute

Create stacks from templates using the console, CLI, or AWS SDK, or create stack sets across multiple accounts and regions



Deploy

Stacks, stack sets, and resources are created, updated, and managed in a uniform way

CloudFormation template

```
1  AWSTemplateFormatVersion: '2010-09-09'
2  > Description: >- ...
8  > Metadata: ...
159 > Parameters: ...
422 > Rules: ...
438 > Conditions: ...
445 > Resources: ...
1469 > Outputs: ...
```

What are the components of a template?

CloudFormation template: Parameters

```
Parameters:
  ADServer1InstanceType:
    AllowedValues:
      - t3.medium
      - t3.large
      - t3.xlarge
      - t3.2xlarge
      - m5.large
      - m5.xlarge
      - m5.2xlarge
      - m5.4xlarge
    Default: m5.large
    Description: Amazon EC2 instance type for the first Active Directory Domain Controller instance
    Type: String
  ADServer1NetBIOSName:
    AllowedPattern: '^[a-zA-Z0-9]+$'
    Default: DC1
    Description: NetBIOS name of the first Active Directory Domain Controller (up to 15 characters)
    MaxLength: '15'
    MinLength: '1'
    Type: String
```

CloudFormation template: Rules

```
Rules:
- SubnetsInVPC:
  Assertions:
  - Assert: !EachMemberIn
    - !ValueOfAll
      - AWS::EC2::Subnet::Id
      - VpcId
      - !RefAll 'AWS::EC2::VPC::Id'
    AssertDescription: All subnets must in the VPC
- S3CRLBucketNameValidation:
  RuleCondition: !And
  - !Equals [!Ref UseS3ForCRL, 'Yes']
  - !Not [!Equals [!Ref PKI, 'No']]
  Assertions:
  - AssertDescription: CRL BucketName cannot must be valid BucketName
  - Assert: !Not [!Equals [!Ref S3CRLBucketName, 'examplebucket']]
```

CloudFormation template: Conditions

Conditions:

- ShouldCreateDHCPOption: !Not [!Equals [!Ref DHCPOptionSet, 'No']]
- ShouldCreateOneTierPkiResource: !Equals [!Ref PKI, 'One-Tier']
- ShouldCreateTwoTierPkiResource: !Equals [!Ref PKI, 'Two-Tier']
- UsingDefaultBucket: !Equals [!Ref QSS3BucketName, 'aws-quickstart']
- AppInsightsEnabled: !Equals [!Ref SetupAppInsightsMonitoring, 'true']
- UsingDefaultAppName: !Equals [!Ref AppInsightsApplicationName, '']

CloudFormation template: Conditions

Conditions:

- ShouldCreateDHCPOption: !Not [!Equals [!Ref DHCPOptionSet, 'No']]
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- AppInsightsEnabled: !Equals [!Ref SetupAppInsightsMonitoring, 'true']
- UsingDefaultAppName: !Equals [!Ref AppInsightsApplicationName, '']

CloudFormation template: Resources

```
- PolicyDocument:
  Version: '2012-10-17'
  Statement:
    - Effect: Allow
      Action: s3:ListBucket
      Resource: !Sub
        - 'arn:${AWS::Partition}:s3:::${S3Bucket}'
        - S3Bucket: !If [UsingDefaultBucket, !Sub '${QSS3BucketName}-${AWS::Region}', !Ref QSS3BucketName]
    - Effect: Allow
      Action: s3:GetObject
      Resource: !Sub
        - 'arn:${AWS::Partition}:s3:::${S3Bucket}/${QSS3KeyPrefix}*'
        - S3Bucket: !If [UsingDefaultBucket, !Sub '${QSS3BucketName}-${AWS::Region}', !Ref QSS3BucketName]
    - Effect: Allow
      Action: ssm:StartAutomationExecution
      Resource: !Sub arn:${AWS::Partition}:ssm:${AWS::Region}:${AWS::AccountId}:automation-definition/${AWSQuickstart
    - Effect: Allow
      Action: ssm:SendCommand
      Resource:
        - !Sub arn:${AWS::Partition}:ssm:${AWS::Region}:*:document/AWS-RunRemoteScript
        - !Sub arn:${AWS::Partition}:ssm:${AWS::Region}:*:document/AWS-RunPowerShellScript
    - Effect: Allow
      Action: ssm:SendCommand
      Resource: !Sub arn:${AWS::Partition}:ec2:${AWS::Region}:${AWS::AccountId}:instance/*
      Condition:
        StringEquals:
          'ssm:ResourceTag/aws:cloudformation:stack-name': !Ref AWS::StackName
```

CloudFormation template: Resources

```
DomainController2:
  CreationPolicy:
    ResourceSignal:
      Timeout: PT60M
      Count: 1
  DependsOn: DomainController1
  Type: AWS::EC2::Instance
  Properties:
    ImageId: !Ref WINFULLBASE
    IamInstanceProfile: !Ref ADServerProfile
    InstanceType: !Ref ADServer2InstanceType
    SubnetId: !Ref PrivateSubnet2ID
    Tags:
      - Key: Name
        Value: !Ref ADServer2NetBIOSName
      - Key: Domain
        Value: !Ref DomainDNSName
      - Key: Role
        Value: Domain Controller
    BlockDeviceMappings:
      - DeviceName: /dev/sda1
        Ebs:
          VolumeSize: 60
          VolumeType: gp3
          Encrypted: true
          KmsKeyId: !Ref EbsEncryptionKmsKeyId
          DeleteOnTermination: true
      - DeviceName: /dev/xvdf
        Ebs:
          VolumeSize: !Ref DataDriveSizeGiB
          VolumeType: gp3
          Encrypted: true
          KmsKeyId: !Ref EbsEncryptionKmsKeyId
          DeleteOnTermination: true
```

CloudFormation template: Resources

```
SecurityGroupIds:
- !Ref DomainControllersSG
PrivateIpAddress: !Ref ADServer2PrivateIP
KeyName: !Ref KeyPairName
UserData:
  Fn::Base64: !Sub
    |
    <powershell>
    $Params = @{
      ADAdminSecParamName = '${ADAdminSecrets}'
      ADServer1NetBIOSName = '${ADServer1NetBIOSName}'
      ADServer1PrivateIP = '${ADServer1PrivateIP}'
      ADServer2NetBIOSName = '${ADServer2NetBIOSName}'
      ADServer2PrivateIP = '${ADServer2PrivateIP}'
      ADAltUserSecParamName = '${ADAltUserSecrets}'
      CreateDefaultOUs = '${CreateDefaultOUs}'
      DeletedObjectLifetime = '${DeletedObjectLifetime}'
      DomainDNSName = '${DomainDNSName}'
      DomainNetBIOSName = '${DomainNetBIOSName}'
      EnableAdvancedAuditingandMetrics = '${EnableAdvancedAuditingandMetrics}'
      RestoreModeSecParamName = '${RestoreModeSecrets}'
      QSS3BucketName = '${QSS3BucketName}'
      QSS3BucketRegion = '${QSS3BucketRegion}'
      QSS3KeyPrefix = '${QSS3KeyPrefix}'
      StackName = '${AWS::StackName}'
      TombstoneLifetime = '${TombstoneLifetime}'
      URLSuffix = '${AWS::URLSuffix}'
      VPCCIDR = '${VPCCIDR}'
    }
    Start-SSMAutomationExecution -DocumentName '${AWSQuickstartActiveDirectoryDS}' -Parameter $Params
    </powershell>
  - QSS3BucketName: !If [UsingDefaultBucket, !Sub '${QSS3BucketName}-${AWS::Region}', !Sub '${QSS3BucketName}']
  - QSS3BucketRegion: !If [UsingDefaultBucket, !Sub '${AWS::Region}', !Sub '${QSS3BucketRegion}']
```

CloudFormation template: Output

```
Outputs:
  · ADSecretsArn:
  · · Description: Alternate AD User Secrets ARN
  · · Value: !Ref ADAltUserSecrets
  · DC1InstanceId:
  · · Description: DomainController 1 instance ID
  · · Value: !Ref DomainController1
  · DC2InstanceId:
  · · Description: DomainController 2 instance ID
  · · Value: !Ref DomainController2
  · DomainAdmin:
  · · Description: Domain administrator account
  · · Value: !Sub ${DomainNetBIOSName}\${DomainAdminUser}
  · DomainControllersSGID:
  · · Description: Domain Controllers Security Group ID
  · · Value: !Ref DomainControllersSG
  · DomainMemberSGID:
  · · Description: Domain Member Security Group ID
  · · Value: !Ref DomainMembersSG
```

Authoring best practices

- *Parameters*: Avoids hardcoding of values, can add validation to users and improve UX with console grouping, labels, and descriptions; keep secrets in AWS Systems Manager Parameter Store and AWS Secrets Manager
- *Mappings*: As a case statement, helps maintain a set of information for various environments
- *Conditions*: Simple if/then statements (e.g., “if dev do this, if prod do that”)
- Imports and exports
- Leverage SSM parameter store for latest AMI instance IDs



YAML/JSON
template

Parameters

Mappings

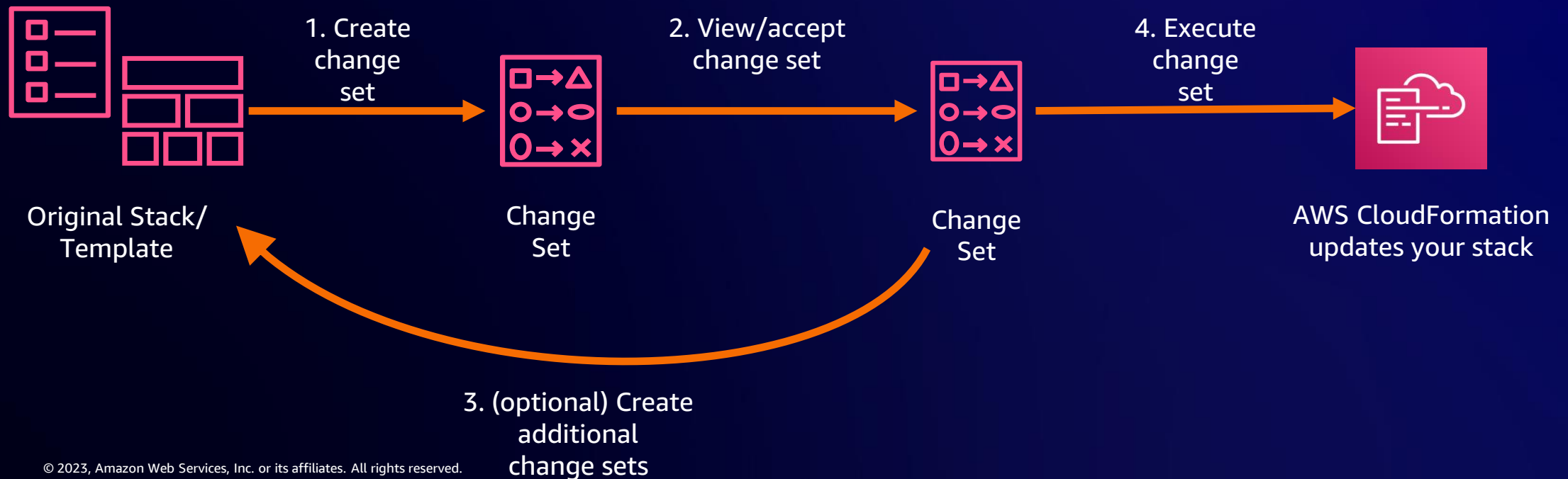
Conditions

Resources

Outputs

AWS CloudFormation – Change Sets

- Change sets enable you to preview how proposed changes to a stack might affect your running resources
- AWS CloudFormation changes your stack only when you decide to execute the change set

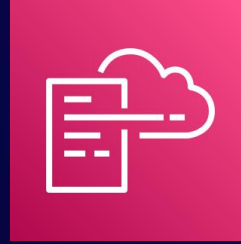


CloudFormation testing



Cfn-lint

- Test YAML/JSON CloudFormation templates against the spec
- Rules are created for best practices
- Rules are extendable to include an organization's best practices



CloudFormation Guard

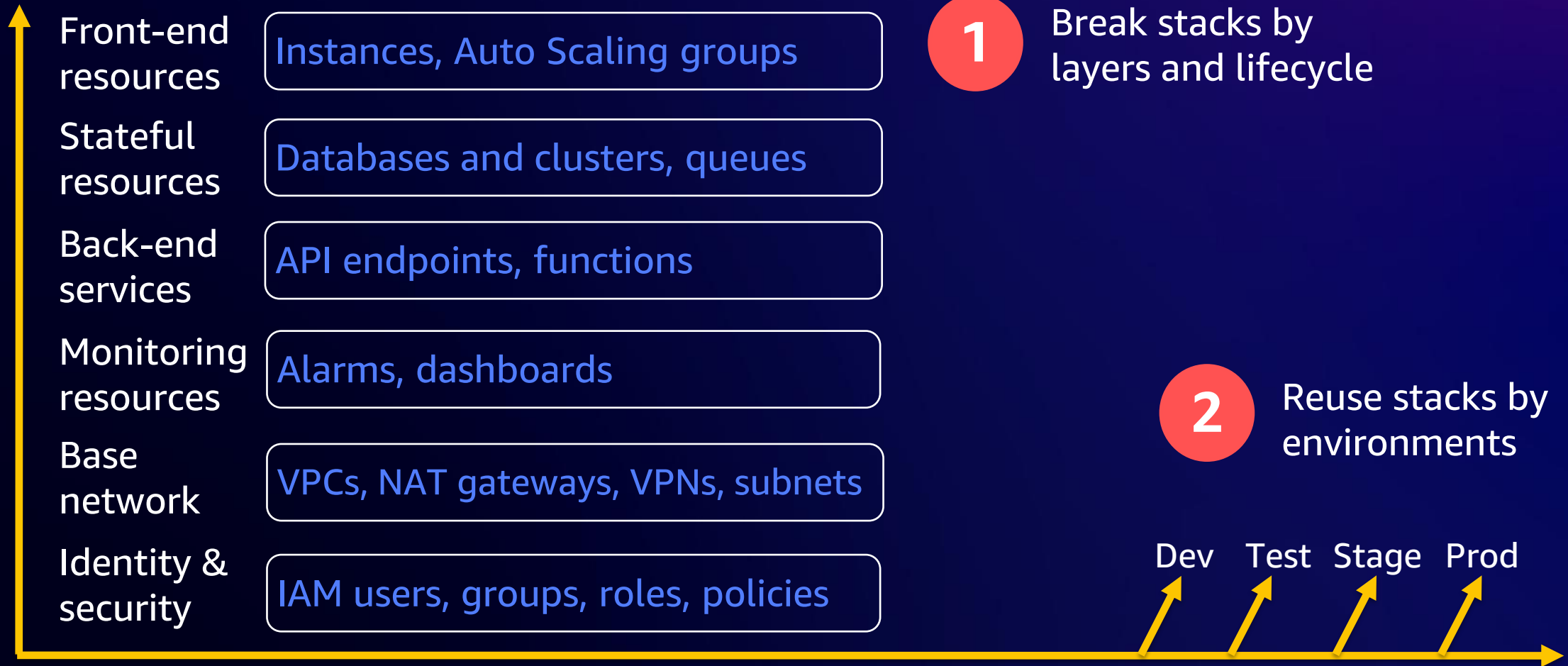
- Validate any YAML/JSON against self-defined rules using the Guard DSL
- Easy to write and create rules to look for an organization's best practices



TaskCat

- Facilitates deploying YAML/JSON templates to many regions
- Provides a report based on the success of the deployment

Stacks by lifecycle



What is on your mind?



Thank you!



Please complete the session survey in the mobile app

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